1.1.52

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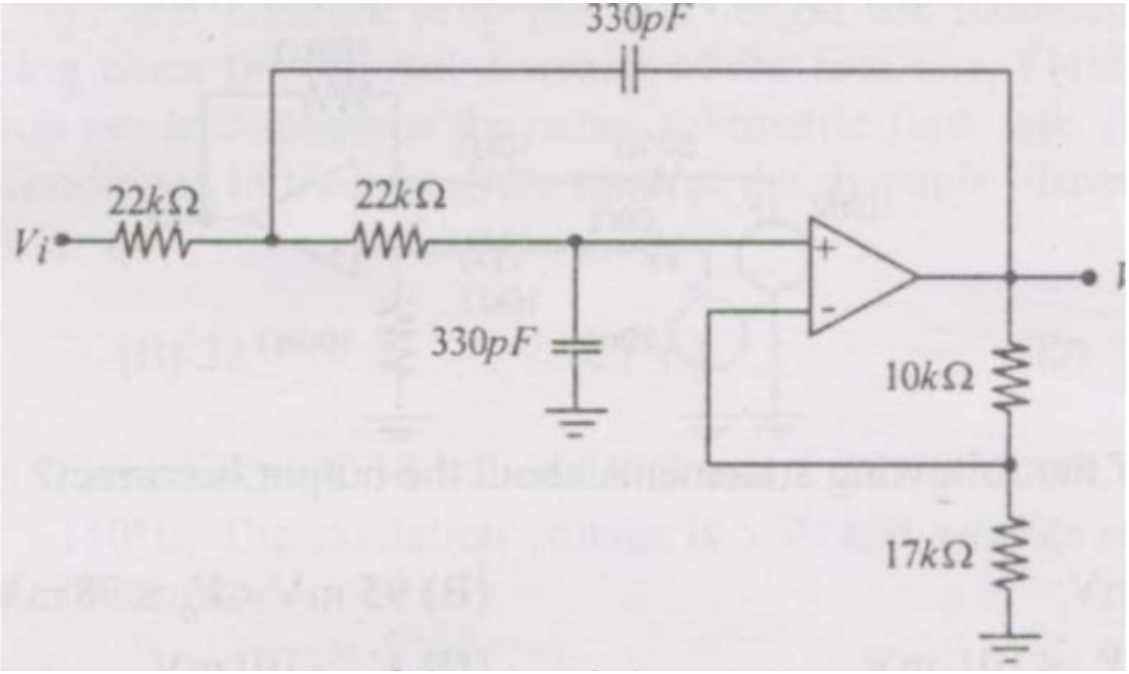
Question

Consider the circuit shown below.



The correct frequency response of the circuit is :

Options

a)

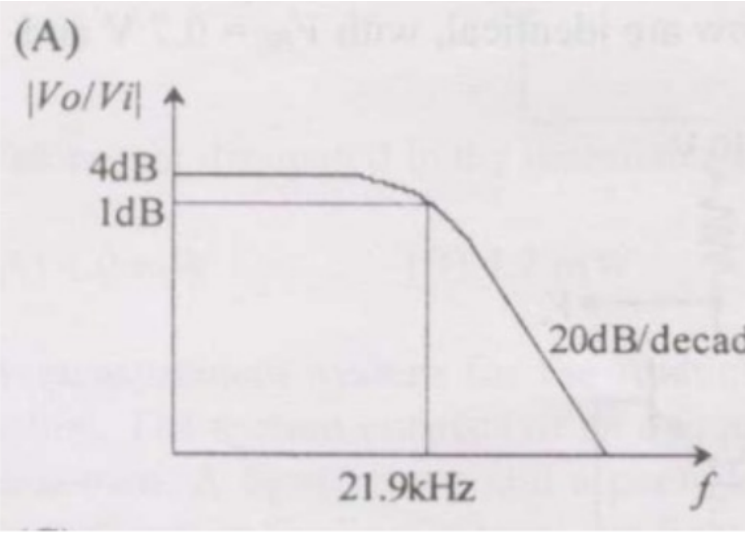


Fig. 1.1.52.2

b)

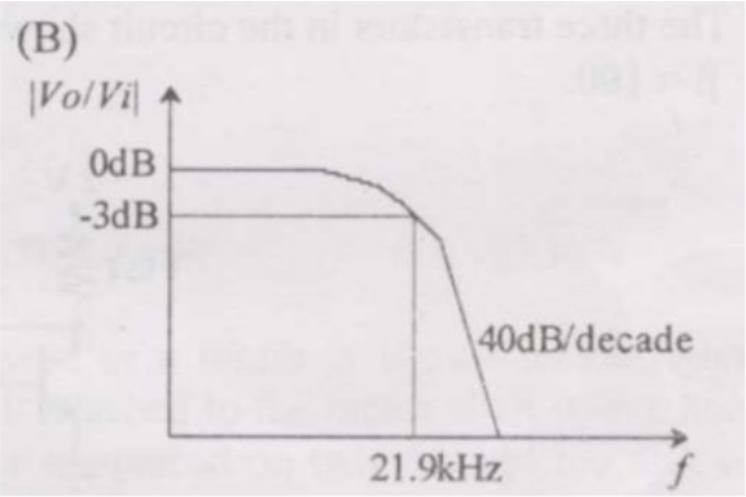


Fig. 1.1.52.3

c)

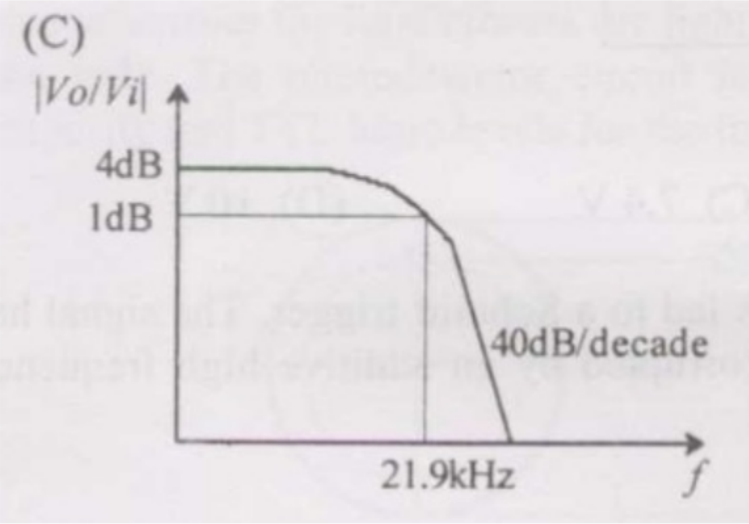


Fig. 1.1.52.4

d)

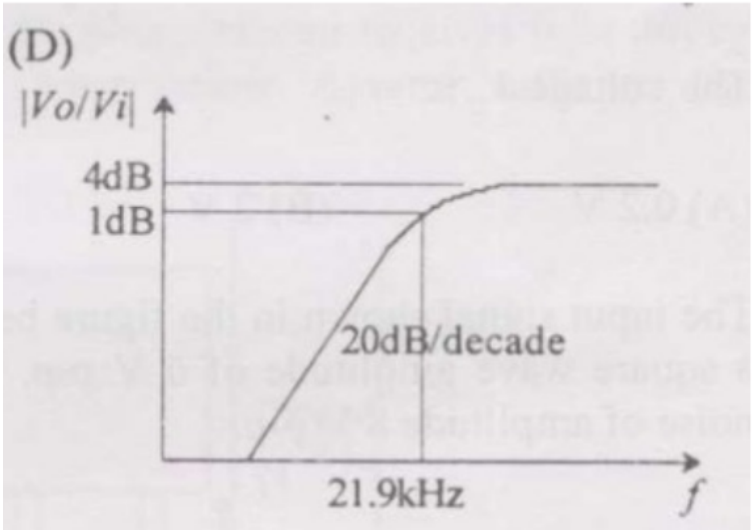


Fig. 1.1.52.5

Step 1: DC Gain K of Op-Amp Stage

The op-amp is wired as a **non-inverting amplifier**:

$$K = 1 + \frac{R_f}{R_g} = 1 + \frac{10\,\mathrm{k}}{17\,\mathrm{k}} = \frac{17+10}{17} = \frac{27}{17}$$

DC gain in decibels

$$K_{\mathrm{dB}} = 20\log_{10}\left(\frac{27}{17}\right) = 20\log_{10}(1.588) \approx \mathbf{4}\,\mathrm{dB}$$

Physical meaning: At very low frequencies, the capacitors are open circuits. No current flows through them, so the RC network passes V_i straight to the op-amp + input. The output is simply $V_o = K \cdot V_i$.

Step 2: KCL at Node B ($= V_+$ input)

Node B is connected to: R_2 (from node A), C_2 (to GND), and op-amp + (draws 0 current).

KCL at B (currents leaving node B):

$$\frac{V_B - V_A}{R} + V_B \cdot sC = 0$$

Rearranging:

$$V_A - V_B = V_B \cdot sRC$$

Equation (1)

$$V_A = V_B (1 + sRC)$$

Interpretation: The voltage at A is always larger than at B by a frequency-dependent factor. At DC ($s = 0$), $V_A = V_B$. At high frequency, the gap grows because C_2 increasingly short-circuits node B to ground.

Step 3: KCL at Node A

Node A is connected to: R_1 (from V_i), R_2 (to B), and C_1 (to V_o).

KCL at A (currents leaving):

$$\frac{V_A - V_i}{R} + \frac{V_A - V_B}{R} + (V_A - V_o) \cdot sC = 0$$

Multiply through by R:

$$(V_A - V_i) + (V_A - V_B) + sRC(V_A - V_o) = 0$$

Collect terms:

Equation (2)

$$V_i = V_A(2 + sRC) - V_B - sRC \cdot V_o$$

Note: The capacitor C_1 creates a **feedback path** from output back to node A. This is what makes the filter 2nd-order (two reactive elements in the signal path).

Step 4: Combine Equations (1) and (2)

Substitute $V_A = V_B(1 + sRC)$ into Equation (2):

$$V_i = V_B(1 + sRC)(2 + sRC) - V_B - sRC \cdot V_o$$

Expand the product:

$$(1 + sRC)(2 + sRC) = 2 + 3sRC + s^2R^2C^2$$

Therefore:

$$V_i = V_B\left(1 + 3sRC + s^2R^2C^2\right) - sRC \cdot V_o$$

Now use $V_B = V_o/K$ (since $V_o = KV_B$):

$$V_i = \frac{V_o}{K}\left(1 + 3sRC + s^2R^2C^2\right) - sRC \cdot V_o$$

$$V_i = V_o\left[\frac{s^2R^2C^2 + sRC(3 - K) + 1}{K}\right]$$

Step 5: Transfer Function

Rearranging for $H(s) = V_o/V_i$:

$$H(s) = \frac{K}{s^2R^2C^2 + sRC(3-K) + 1}$$

Let $\omega_0 = 1/RC$ (so $R^2C^2 = 1/\omega_0^2$). Multiply top and bottom by ω_0^2:

Transfer Function (Standard 2nd-Order LPF)

$$\boxed{H(s) = \frac{K\,\omega_0^2}{s^2 + \omega_0(3-K)\,s + \omega_0^2}}$$

Comparing with $H(s) = \dfrac{K\omega_0^2}{s^2 + (\omega_0/Q)\,s + \omega_0^2}$:

$$\frac{\omega_0}{Q} = \omega_0(3-K) \quad \Longrightarrow \quad Q = \frac{1}{3-K}$$

Key insight: The gain K controls *both* the passband gain *and* the damping. Increasing K raises both gain and Q.

Step 6: Corner Frequency f_0

$$\omega_0 = \frac{1}{RC} = \frac{1}{22 \times 10^3 \times 330 \times 10^{-12}} = \frac{1}{7.26 \times 10^{-6}}$$

$$\omega_0 = 1.377 \times 10^5 \text{ rad/s}$$

Corner Frequency

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1.377 \times 10^5}{6.283} \approx \mathbf{21.9}\,\text{kHz}$$

Physical meaning: f_0 is the frequency at which the reactive impedance $1/\omega_0 C$ equals the resistance R. Below f_0: resistors dominate, gain is flat. Above f_0: capacitors dominate, gain falls.

Step 7: Quality Factor Q

$$3 - K = 3 - \frac{27}{17} = \frac{51 - 27}{17} = \frac{24}{17}$$

$$Q = \frac{1}{3 - K} = \frac{17}{24} = 0.7083 \approx \frac{1}{\sqrt{2}}$$

Butterworth Condition: $Q = 1/\sqrt{2}$

A filter with $Q = 1/\sqrt{2}$ is called a **Butterworth filter**. It has the *maximally flat magnitude response* in the passband: no peaking, no ripple — the smoothest possible roll-off.

Effect of Q on shape:

- $Q < 1/\sqrt{2}$: overdamped, sluggish roll-off
- $Q = 1/\sqrt{2}$: Butterworth, flat then smooth fall
- $Q > 1$: underdamped, peak appears near f_0

Step 8: Gain at Corner Frequency

Substitute $s = j\omega_0$ into $H(s)$:

$$H(j\omega_0) = \frac{K\omega_0^2}{(j\omega_0)^2 + \omega_0(3-K)(j\omega_0) + \omega_0^2}$$

$$= \frac{K\omega_0^2}{-\omega_0^2 + j\omega_0^2(3-K) + \omega_0^2} = \frac{K}{j(3-K)}$$

$$|H(j\omega_0)| = \frac{K}{3-K} = \frac{27/17}{24/17} = \frac{27}{24} = 1.125$$

Gain at f_0

$$20\log_{10}(1.125) \approx \mathbf{1}\,\text{dB} = K_{\text{dB}} - 3\,\text{dB}$$

This is a universal property of Butterworth filters: the gain is always exactly $-3\,\text{dB}$ below the passband gain at the corner frequency.

Step 9: Roll-off Rate (Order of Filter)

The denominator of $H(s)$ is a **degree-2 polynomial in** s.

At very high frequencies ($\omega \gg \omega_0$), the s^2 term dominates:

$$|H(j\omega)|\Big|_{\omega \gg \omega_0} \approx \frac{K\omega_0^2}{\omega^2} \propto \frac{1}{\omega^2} = \frac{1}{(2\pi f)^2}$$

Doubling frequency ($+1$ decade $= \times 10$): gain drops by factor $10^2 = 100$:

$$20\log_{10}(1/100) = -40\,\text{dB per decade}$$

Roll-off Rule

$$\text{Roll-off rate} = -20\,\text{dB/decade} \times \text{order}$$

2nd order $\Rightarrow$ $-\mathbf{40}$ dB/decade

Eliminating Wrong Options

Option (A) — Wrong

4 dB, 1 dB at 21.9 kHz,
20 dB/decade roll-off.

20 dB/decade $\Rightarrow$ 1st order.
A 1st-order filter needs only *one* reactive element.
This circuit has two capacitors $\Rightarrow$ must be 2nd order. $\times$

Option (B) — Wrong

0 dB DC gain, -3 dB at corner,
40 dB/decade.

0 dB $\Rightarrow K = 1$.
$K = 1$ requires $R_f = 0$ (voltage follower).
Here $R_f = 10\,\text{k}\Omega \neq 0$. $\times$

Option (D) — Wrong

High-pass shape (gain rises to 4 dB at 21.9 kHz).

A high-pass passes high f, blocks low f. Our circuit passes DC and blocks high

Option (C) — Correct ✓

4 dB DC gain,
drops to 1 dB at 21.9 kHz,
40 dB/decade roll-off,
no peaking (Butterworth).

All four computed properties

Plot

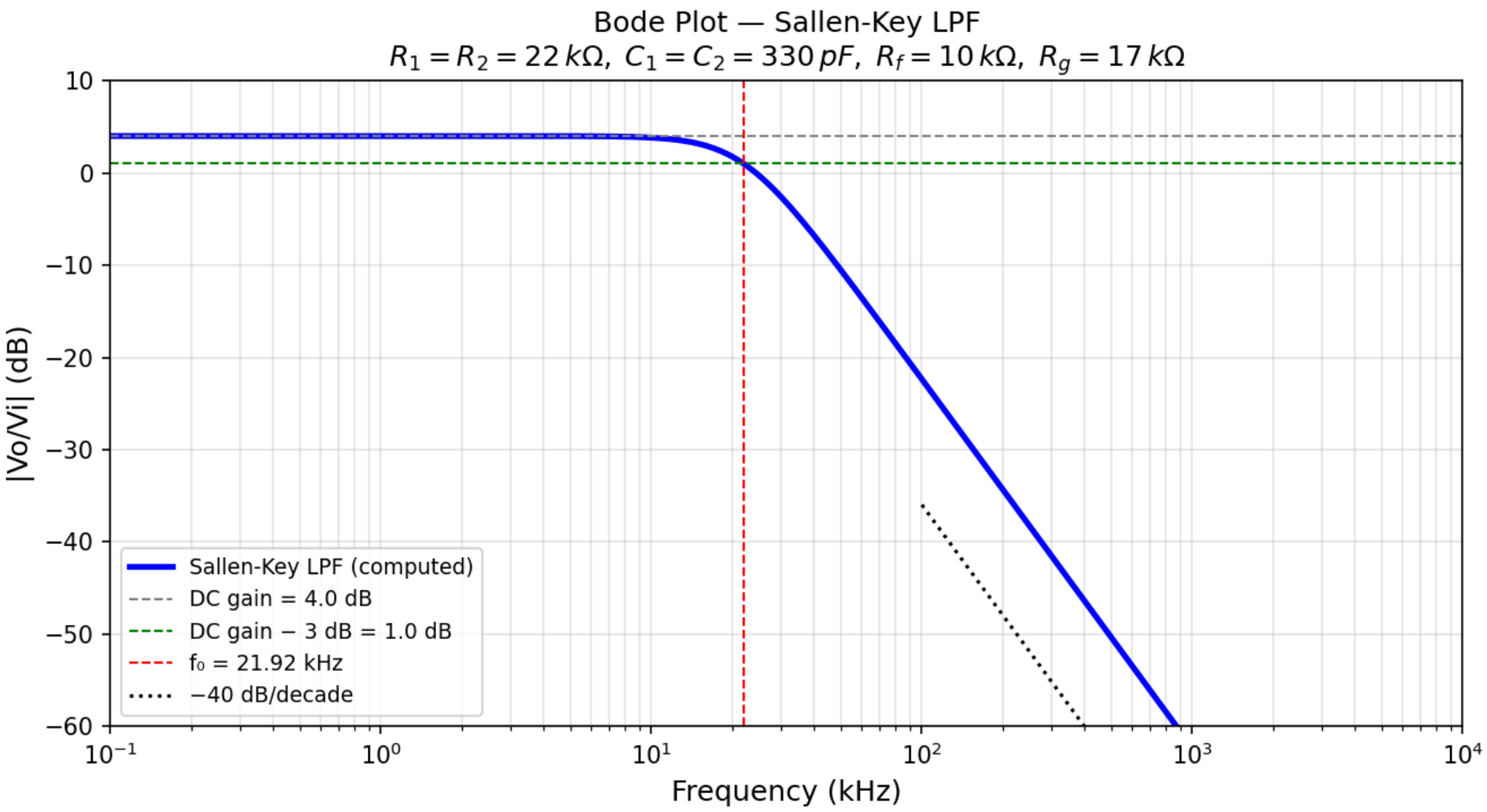

Bode Plot — Sallen-Key LPF
$R_1 = R_2 = 22\,k\Omega,\ C_1 = C_2 = 330\,pF,\ R_f = 10\,k\Omega,\ R_g = 17\,k\Omega$
|Vo/Vi| (dB)
Frequency (kHz)
Sallen-Key LPF (computed)
DC gain = 4.0 dB
DC gain − 3 dB = 1.0 dB
f₀ = 21.92 kHz
−40 dB/decade

Figure:

Python Verification — Parameters & Transfer Function

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

# Parameters
R  = 22e3;   C  = 330e-12     # Ohm, F
Rf = 10e3;   Rg = 17e3        # Ohm

K  = 1 + Rf / Rg
w0 = 1 / (R * C)
f0 = w0 / (2 * np.pi)
Q  = 1 / (3 - K)

print(f"K      = {K:.6f}   ({20*np.log10(K):.2f} dB)")
print(f"f0     = {f0/1e3:.3f} kHz")
print(f"omega0 = {w0:.2f} rad/s")
print(f"Q      = {Q:.4f}   (1/sqrt(2) = {1/np.sqrt(2):.4f})")
print(f"Filter : {'Butterworth' if abs(Q - 1/np.sqrt(2)) < 0.01 else '
     Other'}")

# Transfer function: H(s) = K*w0^2 / (s^2 + w0*(3-K)*s + w0^2)
num = [K * w0**2]
den = [1, w0 * (3 - K), w0**2]
```

Python Verification — Bode Plot

```
f = np.logspace(2, 7, 2000)
w = 2 * np.pi * f
_, H = signal.freqs(num, den, worN=w)
mag_dB = 20 * np.log10(np.abs(H))

fig, ax = plt.subplots(figsize=(9, 5))
ax.semilogx(f / 1e3, mag_dB, 'b-', lw=2.5, label='Sallen-Key␣LPF␣(
    computed)')
ax.axhline(20*np.log10(K),     color='gray',  ls='--', lw=1, label=f'DC␣
    gain␣=␣{20*np.log10(K):.1f}␣dB')
ax.axhline(20*np.log10(K) - 3, color='green', ls='--', lw=1, label=f'DC␣
    gain␣-␣3␣dB␣=␣{20*np.log10(K)-3:.1f}␣dB')
ax.axvline(f0 / 1e3,           color='red',   ls='--', lw=1, label=f'f0␣=
    ␣{f0/1e3:.2f}␣kHz')

f_slope = np.array([1e5, 1e6])
ref_dB  = 20 * np.log10(K) - 40
ax.semilogx(f_slope/1e3, [ref_dB, ref_dB-40], 'k:', lw=1.5, label='-40␣dB
    /decade')

ax.set_xlabel('Frequency␣(kHz)');  ax.set_ylabel('|Vo/Vi|␣(dB)')
ax.set_xlim([0.1, 1e4]);           ax.set_ylim([-60, 10])
ax.legend(fontsize=9);             ax.grid(True, which='both', alpha
    =0.35)
plt.tight_layout();                plt.savefig('bode_1_1_52.png', dpi
    =150)

# Verify gain at f0
_, H_at_f0 = signal.freqs(num, den, worN=[w0])
print(f"Gain␣at␣f0:␣{20*np.log10(abs(H_at_f0[0])):.3f}␣dB␣␣"
f"(expected␣{20*np.log10(K)-3:.3f}␣dB)")
```